LECTURE ON BASICS
Dimensions, Units and Dimensional Homogeneity

Reference: Munson et al., Fundamentals of Fluid Mechanics, Chapter 1

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Dimensions and units

A qualitative description identifies the nature of a quantity: length, time, mass, temperature. These are the primary quantities, and every other quantity is secondary, with dimensions that follow from them.

$$A \doteq L^2, \qquad v \doteq LT^{-1}, \qquad \rho \doteq ML^{-3}, \qquad V \doteq L^3$$

Two systems are in use: the mass–length–time system (MLT) and the force–length–time system (FLT). Newton's second law links them:

$$F \doteq MLT^{-2}, \qquad M \doteq FL^{-1}T^2$$

A quantitative description also needs a unit: the metre, the second, the kilogram. The symbol ≐ means "has the dimensions of".

Dimensions of common quantities

Table 1.1

Dimensions Associated with Common Physical Quantities

	FLT System	*MLT* System
Acceleration	LT^{-2}	LT^{-2}
Angle	$F^0L^0T^0$	$M^0L^0T^0$
Angular acceleration	T^{-2}	T^{-2}
Angular velocity	T^{-1}	T^{-1}
Area	L^2	L^2
Density	$FL^{-4}T^2$	ML^{-3}
Energy	FL	ML^2T^{-2}
Force	F	MLT^{-2}
Frequency	T^{-1}	T^{-1}
Heat	FL	ML^2T^{-2}
Length	L	L
Mass	$FL^{-1}T^2$	M
Modulus of elasticity	FL^{-2}	$ML^{-1}T^{-2}$
Moment of a force	FL	ML^2T^{-2}
Moment of inertia (area)	L^4	L^4
Moment of inertia (mass)	FLT^2	ML^2
Momentum	FT	MLT^{-1}
Power	FLT^{-1}	ML^2T^{-3}
Pressure	FL^{-2}	$ML^{-1}T^{-2}$
Specific heat	$L^2T^{-2}\Theta^{-1}$	$L^2T^{-2}\Theta^{-1}$
Specific weight	FL^{-3}	$ML^{-2}T^{-2}$
Strain	$F^0L^0T^0$	$M^0L^0T^0$
Stress	FL^{-2}	$ML^{-1}T^{-2}$
Surface tension	FL^{-1}	MT^{-2}
Temperature	Θ	Θ
Time	T	T
Torque	FL	ML^2T^{-2}
Velocity	LT^{-1}	LT^{-1}
Viscosity (dynamic)	$FL^{-2}T$	$ML^{-1}T^{-1}$
Viscosity (kinematic)	L^2T^{-1}	L^2T^{-1}
Volume	L^3	L^3
Work	FL	ML^2T^{-2}

Dimensional homogeneity

All theoretically derived equations are dimensionally homogeneous: every additive term has the same dimensions.

$$v = v_0 + a\,t$$

$$LT^{-1} \doteq LT^{-1} + LT^{-2}\,T$$

Homogeneous, and valid in any consistent system of units: a general homogeneous equation.

$$d = 4.91\,t^2$$

$$d = \frac{g\,t^2}{2}$$

The constant 4.91 carries the dimensions LT^{-2}, so this form is valid only in metres and seconds: a restricted homogeneous equation.

Always prefer the general form

Systems of units

A quantity needs a unit for each basic dimension: length, mass, time and temperature. This course works entirely in the International System (SI): metre, kilogram, second, kelvin.

$$1\ \mathrm{N} = (1\ \mathrm{kg})(1\ \mathrm{m/s^2}), \qquad 1\ \mathrm{Pa} = 1\ \mathrm{N/m^2} \qquad g = 9.807\ \mathrm{m/s^2}$$

$$\mathrm{K} = {}^\circ\mathrm{C} + 273.15$$

and 1 J = 1 N·m, 1 W = 1 J/s.

Two older systems still appear in the literature and in the textbook: British Gravitational (ft, slug, s, °R) and English Engineering (ft, lbm, lb, s, °R). Data given in either must be converted before anything is substituted into an equation.

Prefixes for SI units

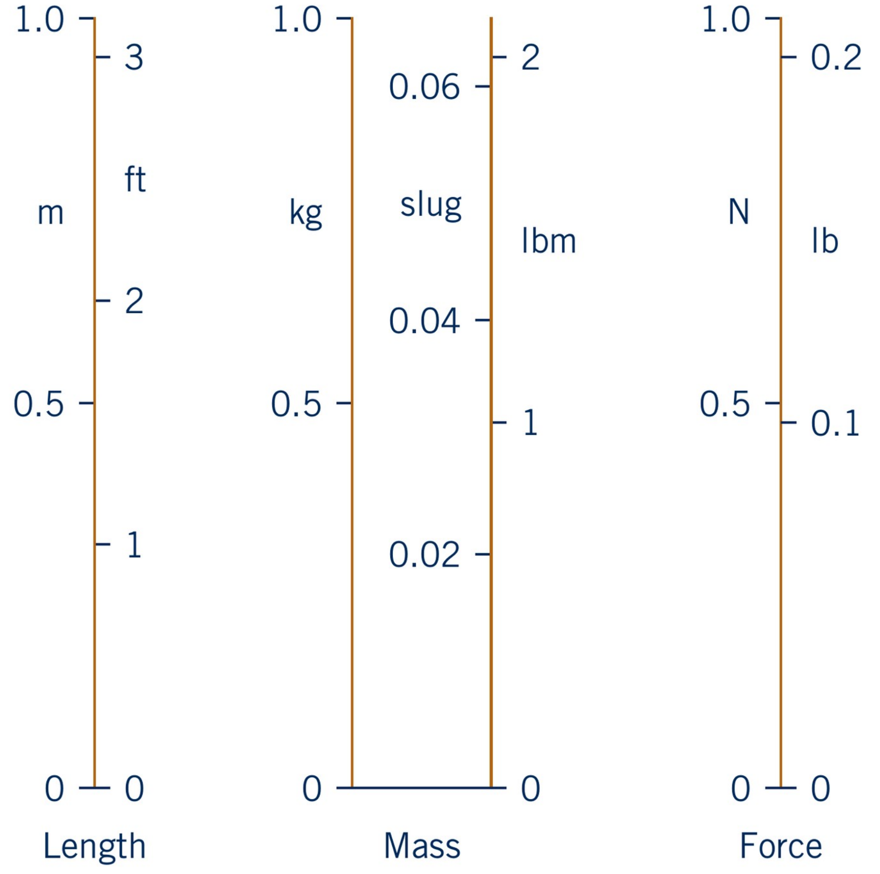
Table 1.2

Prefixes for SI Units

Factor by Which Unit Is Multiplied	Prefix	Symbol	Factor by Which Unit Is Multiplied	Prefix	Symbol
10^{15}	peta	P	10^{-2}	centi	c
10^{12}	tera	T	10^{-3}	milli	m
10^{9}	giga	G	10^{-6}	micro	μ
10^{6}	mega	M	10^{-9}	nano	n
10^{3}	kilo	k	10^{-12}	pico	p
10^{2}	hecto	h	10^{-15}	femto	f
10	deka	da	10^{-18}	atto	a
10^{-1}	deci	d			

kN is read as kilonewtons and stands for 10^3 N; mm stands for 10^{-3} m. The centimetre is not an accepted unit of length in fluid mechanics: use millimetres or metres.

Other systems of units



- The three systems use different units for length, mass and force
- Weight and mass are different quantities: $W = mg$
- 1 ft = 0.3048 m, 1 slug = 14.594 kg, 1 lb = 4.448 N
- Convert the data first, then work in SI throughout

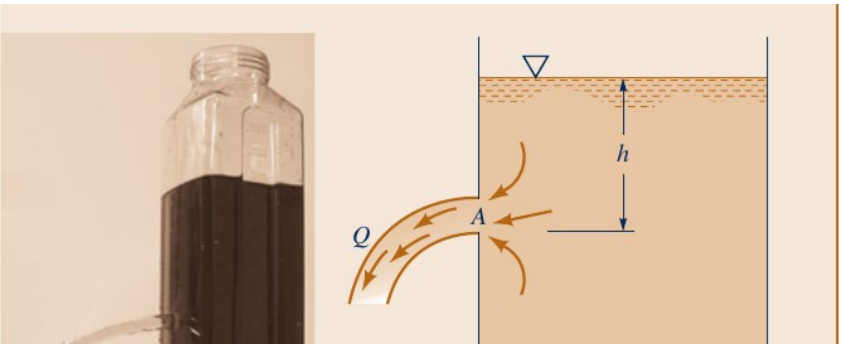
Examples

Example — flow through an orifice

A commonly used equation for the volume rate of flow Q through a sharp-edged orifice in the side of a tank is

$$Q = 0.61\,A\,\sqrt{2\,g\,h}$$

with A the area of the orifice, h the height of the liquid surface above it and g the acceleration of gravity. Show that the equation is dimensionally homogeneous, and find the restricted form it takes when g is replaced by its value in SI units.



Checking the dimensions of both sides:

$$L^3T^{-1} \doteq (L^2)(LT^{-2})^{1/2}(L)^{1/2}$$

The check is satisfied and 0.61√2 is dimensionless, so the equation holds in any consistent system of units.

Substituting g = 9.81 m/s² gives a coefficient that carries units, valid only in metres and seconds:

$$Q = 2.70\,A\,\sqrt{h}$$

Example — force on a support

A tank of liquid with a total mass of 36 kg rests on a support in a space vehicle. The vehicle is accelerating upward at 4.57 m/s². Determine the force that the tank exerts on the support.

The support must both carry the weight of the tank and provide its acceleration, so Newton's second law gives

$$F = m\,(g + a)$$

$$F = 36\,(9.81 + 4.57) = 518\ \mathrm{N}$$

By Newton's third law the tank pushes on the support with 518 N, directed downward.

Example — compressed air tank

A closed tank of volume 0.025 m³ is filled with compressed air at a gage pressure of 350 kPa and a temperature of 20 °C. Determine the density of the air and the weight of the air in the tank. Take R = 286.9 J/(kg·K) for air.

The gas law needs absolute pressure and absolute temperature, so convert both first:

$$p_{\mathrm{abs}} = 350 + 101.3 = 451.3\ \mathrm{kPa}$$

$$\rho = \frac{451.3 \times 10^3\ \mathrm{Pa}}{(286.9)(293.15)} = 5.37\ \mathrm{kg/m^3}$$

$$\mathcal{W} = \rho\, g\, V = (5.37)(9.81)(0.025) = 1.32\ \mathrm{N}$$

With T = 20 + 273.15 = 293.15 K. The mass of air in the tank is ρV = 0.134 kg.

Example — compressed air tank, continued

Repeating the calculation over a range of gage pressures gives the weight of air in the same tank:

The result is a straight line, but it does not pass through the origin: it crosses zero at p = −101.3 kPa (gage), a perfect vacuum.

Doubling the gage pressure therefore does not double the mass of air in the tank.